Ex. No: 4	Demonstration of Email Phishing Attack and Preventive Measures
Date:	

Aim:

To understand the concept of email phishing attacks, analyze phishing email characteristics, and study preventive and mitigation techniques.

- To identify features of phishing emails.
- To demonstrate how phishing attacks are executed (in a controlled environment).
- To implement preventive mechanisms against phishing attacks.

Tools Required

- Isolated lab system / Virtual Machine
- Sample phishing email templates (for educational purpose)
- Email header analyzer tool
- Web browser
- Spam filter settings (Gmail/Outlook demo account)

Theory

Phishing is a social engineering attack where attackers impersonate legitimate organizations to steal sensitive information such as login credentials, banking details, or OTPs.

Common phishing techniques include:

- Email spoofing
- Fake login pages
- Malicious attachments
- Urgent or threatening language

Procedure:

Part A: Demonstration of Phishing Email (Controlled Simulation)

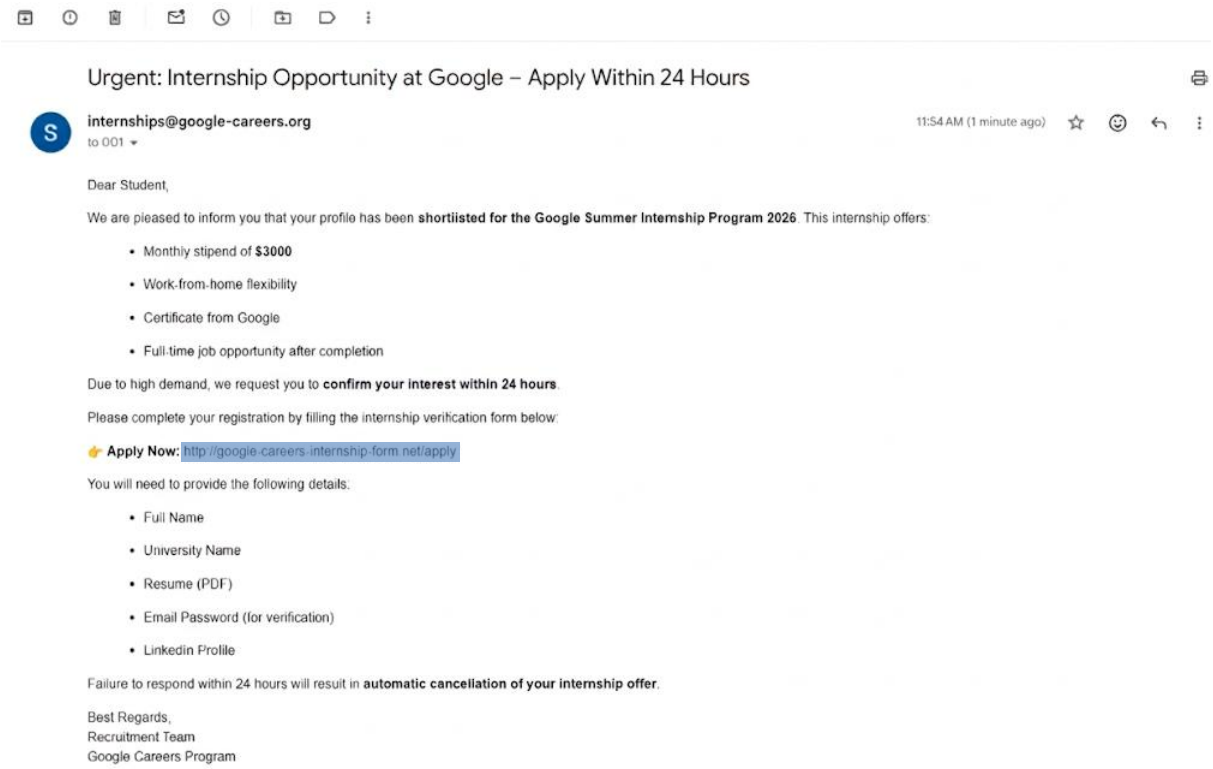
1. Create a sample phishing email template (educational purpose only).
2. Include common phishing characteristics:

    - Fake sender address (e.g., internships@google-careers.org)
    - Urgent subject line ("Urgent: Internship Opportunity at Google – Apply Within 24 Hours")
    - Suspicious link (e.g., http://google-careers-internship-form.net/apply)

3. Analyze the email header using an email header analyzer.

4. Identify red flags:

    - Mismatched/Deceptive Domain (e.g., google-careers.org instead of https://www.google.com/search?q=google.com)
    - Requests for Sensitive Information (e.g., asking for an "Email Password")
    - Too-Good-To-Be-True Offers (e.g., guaranteed $3000 stipend with no interview)
    - Artificial Urgency (e.g., threatening cancellation within 24 hours)



Urgent: Internship Opportunity at Google – Apply Within 24 Hours

S internships@google-careers.org 11:54 AM (1 minute ago)
to 001

Dear Student,

We are pleased to inform you that your profile has been **shortlisted for the Google Summer Internship Program 2026**. This internship offers:

- Monthly stipend of **$3000**
- Work-from-home flexibility
- Certificate from Google
- Full-time job opportunity after completion

Due to high demand, we request you to **confirm your interest within 24 hours**.

Please complete your registration by filling the internship verification form below:

👉 **Apply Now:** http://google-careers-internship-form.net/apply

You will need to provide the following details:

- Full Name
- University Name
- Resume (PDF)
- Email Password (for verification)
- Linkedin Profile

Failure to respond within 24 hours will result in **automatic cancellation of your internship offer**.

Best Regards,
Recruitment Team
Google Careers Program

Part B: Preventive Measures

1. Enable Spam Filters in email settings.
2. Verify sender domain before clicking links.
3. Check URL by hovering over hyperlinks.
4. Enable Two-Factor Authentication (2FA).
5. Install Anti-Phishing browser extensions.
6. Report phishing emails to the service provider.
7. Educate users on recognizing phishing attempts

Fig 4.1 Turn on spam filtering in email settings

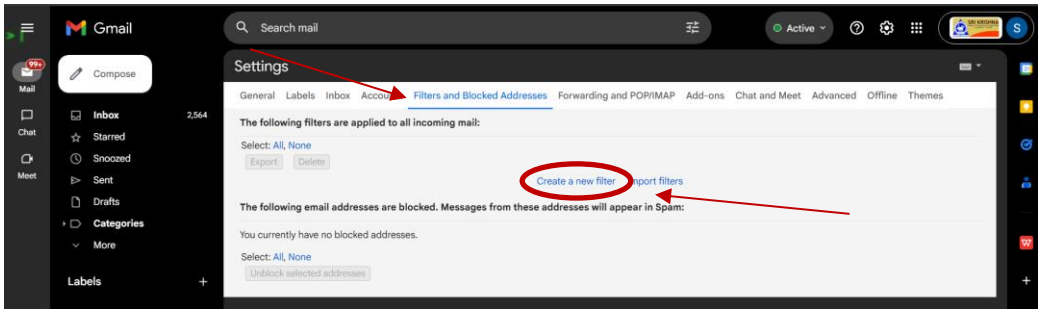


Fig 4.2 Verify sender domain before clicking links.

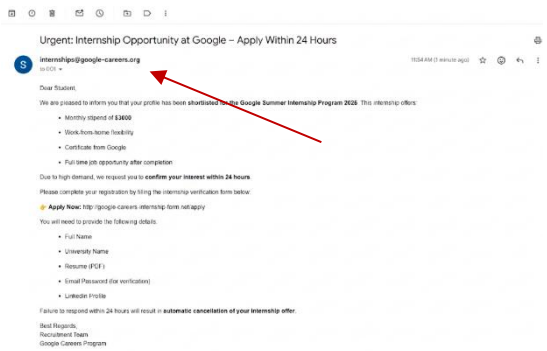


Fig 4.3 Check URL by hovering over hyperlinks.

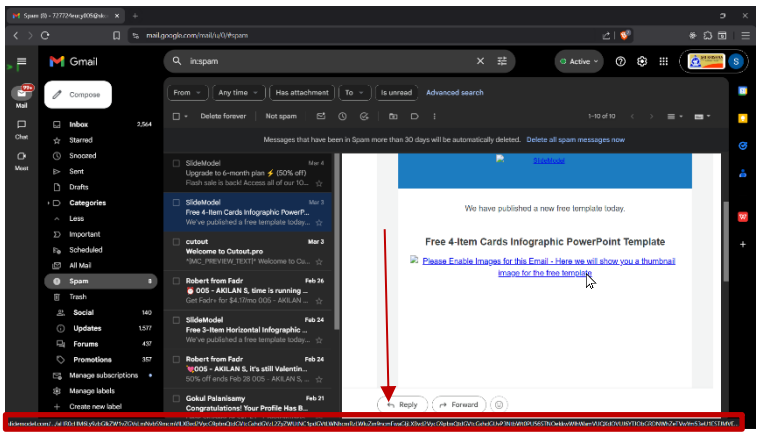


Fig 4.4 Enable Two-Factor Authentication (2FA).

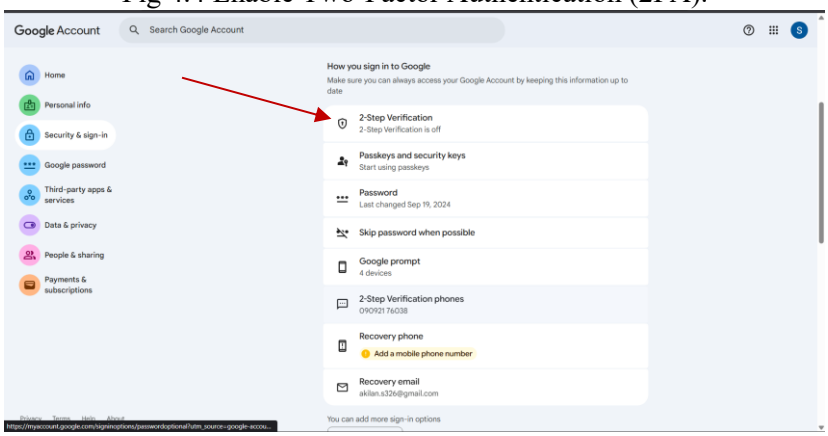


Fig 4.5 Install Anti-Phishing browser extensions.

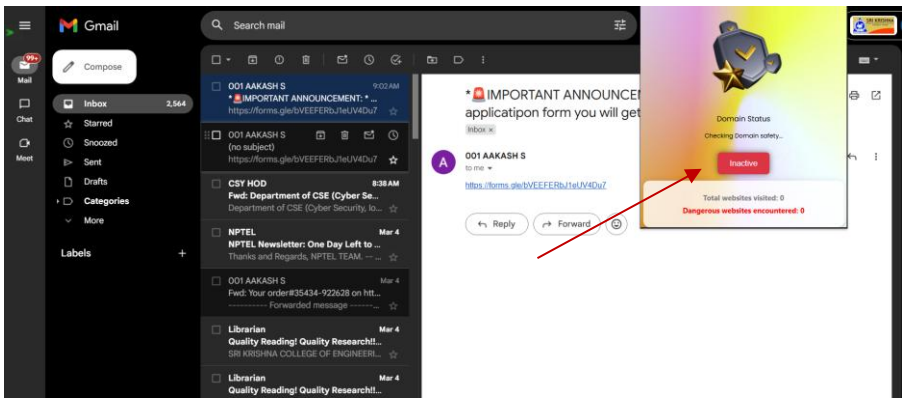
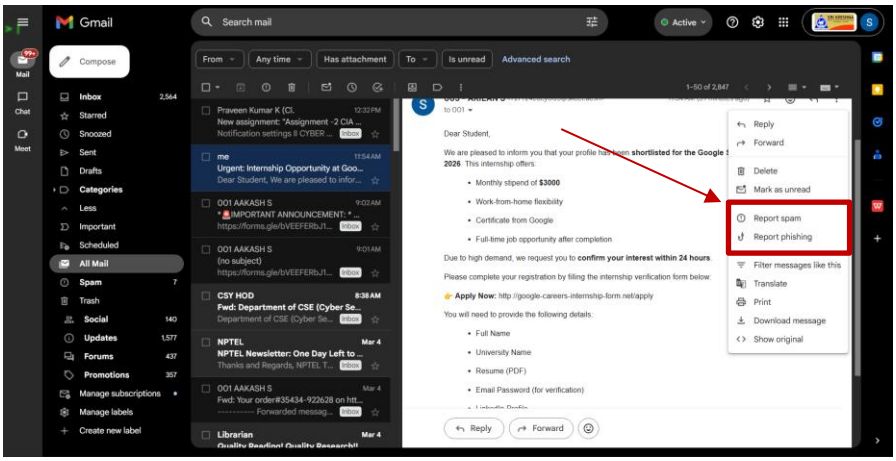


Fig 4.6 Report phishing emails to the service provider.



Sample Phishing Indicators

Indicator	Description
Urgency	“Act within 24 hours”
Suspicious Link	URL different from displayed text
Generic Greeting	“Dear Customer”
Unexpected Attachment	.exe, .zip, .html files

Result:

The phishing email characteristics were successfully identified, and preventive measures were implemented to mitigate phishing risks.